Spatio-Temporal Dual-Branch Network With Predictive Feature Learning for Satellite Video Object Segmentation

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***Abstract*— Satellite video is an important new earth observation data source that can be used to acquire large-scale dynamic information. Satellite video object segmentation (SVOS) is aimed at separating the foreground and background of satellite video and is a fundamental processing task for satellite video. To date, the state-of-the-art research into SVOS has mainly focused on unsupervised target extraction methods through hand-crafted features and post-processing operations, which are prone to obtaining incomplete contours of the targets and result in foreground aperture problem. Furthermore, the small size of targets and appearance deformation make the SVOS more difficult. Therefore, in this article, a spatio-temporal dual-branch network is proposed with predictive feature learning for the SVOS task. The proposed model consists of a temporal coherence branch and a spatial segmentation branch. In the temporal coherence branch, the Wasserstein generative adversarial network (WGAN) architecture is utilized for the future frame prediction to exploit temporal information, which captures the dynamic appearance and motion cues from the unlabeled satellite video data through predictive feature learning module in an adversarial manner. As a result, the proposed method can obtain segmentation results with temporal consistency, while avoiding the generation of optical flow images. In the spatial segmentation branch, a fully convolutional network (FCN) is used to extract the high-level spatial information of the satellite video and achieve end-to-end SVOS, without any post-processing operations. In the network implementation, boundary loss is used to solve the highly unbalanced segmentation problem caused by small size of the targets. The two branches of the network are also mutually constrained, to improve the final object segmentation results. The visual and quantitative results of three experiments all demonstrate that the proposed method outperforms the other current SVOS models.**

***Index Terms*— Generative adversarial network (GAN), predictive feature learning, satellite video, video object segmentation.**

I. Introduction

With the continuous development of remote-sensing satellites, earth observation technology is constantly being enriched. Satellite video is a new type of data source that can provide abundant dynamic information about the earth's surface. The SkySat-1 satellite launched by the United States is the first civilian video satellite that can film a panchromatic video with a duration of up to 90 s. In addition, China has launched the Jilin-1 video satellite constellation, which is the first video satellite constellation and the largest civilian constellation in China. As a result of its real-time remote-sensing imagery acquisition capability, satellite video is useful for monitoring many rapidly changing scenarios [1], such as traffic monitoring [2], [3], military security, and maritime monitoring [4].

In the research into satellite video, video object segmentation and object tracking [5], [6] are the two main tasks for satellite video. This article focuses on the object segmentation task for satellite video. Satellite video object segmentation (SVOS) involves separating a particular object from the background in a satellite video. SVOS is mainly aimed at the key objects, significant objects, and moving objects in the satellite video. Similar to video object segmentation in surveillance video, SVOS plays an important role in object tracking and the further applications of satellite video.

Differing from surveillance video, satellite video involves a higher imaging height and wider imaging range. The spatial resolution of satellite videos is mostly about 1 m, which means that the targets in satellite video are usually of a small size and can be difficult to distinguish. Fig. 1 depicts the two major problems in SVOS. As shown in Fig. 1(a), the appearance of a moving object can change at different moments. In this case, the temporal information is very important for maintaining the consistency of the video object segmentation results. However, as a result of the large amount of data and low spatial resolution in satellite video, extracting the temporal information from satellite video accurately is a challenging task. In addition, the unbalanced foreground and background can be another problem in SVOS. As shown in Fig. 1(b), the size of the target is very small, while the imaging range of the satellite video is very large. As a result, the numbers of pixels in the foreground and background usually differ by several orders of magnitude, which causes a large imbalance between the foreground and background in SVOS.

The state-of-the-art research into SVOS has mainly addressed the moving vehicle extraction task and traffic monitoring. The foreground extraction models in satellite video mainly focus on an unsupervised approach to obtain the

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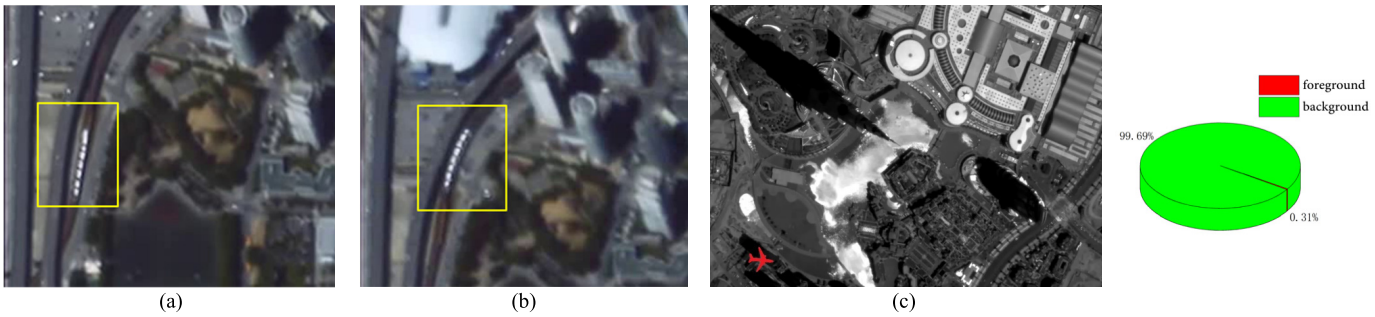


Fig. 1. Two major challenges in satellite video object segmentation. (a) and (b) Adjacent sequences of two moments in a satellite video sequence, respectively, showing the appearance change at the different moments caused by the target's motion. (c) Highly unbalanced foreground and background in the satellite video sequence, in which the marked red area is the foreground while the unmarked area is the background.

moving targets. Examples of such models include optical flow models [7], background subtraction models [8], and frame difference models [9]. The background subtraction models separate the foreground and acquire the moving targets by background modeling and subsequent post-processing operations. Due to its high computational efficiency, background subtraction is widely used in moving object detection for satellite video. However, the moving object detection algorithms are heavily dependent on the quality of the background model. Typical background models include the average background model [10], the Gaussian background model [11], ViBe [12], [13], the codebook background model [14], [15], and robust principal component analysis (RPCA) [16]. However, most background subtraction methods are modeled in such a way that the background is considered to be static, which is contrary to the situations where the background scenarios include many non-stationary scenes. As a result, the complex environment and background movement often result in the background subtraction models acquiring a mass of false alarms in the moving object extraction results.

To eliminate the large amount of false alarms introduced by noise, environmental change, and background displacement in satellite video, most of the state-of-the-art algorithms acquire precise targets by post-processing operations and discrimination processes. Other algorithms use a manually drafted mask to filter the noise interference in the background area. To reduce the need for manual intervention in the process, the motion area can be acquired by a motion heat map through motion segmentation and trajectory accumulation [2]. A velocity threshold can also be used to eliminate the influence of background displacement [17]. Motion flow clustering is an efficient way to gather motion patterns and filter falsely detected candidates which have different motion patterns [18]. However, the above algorithms require a large amount of human effort, making the processing more complicated. In addition, the foreground results acquired by the subtraction operation may contain some noise and objects that are not of interest. To solve this problem, morphological operations are generally used to remove the very small and very large objects, and the motion trajectory length [17] is applied to filter the noise. Although it achieves a good performance in removing nonmoving targets, it may cause partial error filtering and missing targets due to the lack of valid supervision information, and it cannot obtain end-to-end video object segmentation results. What is more, the segmentation results acquired by the background subtraction methods are prone to causing foreground aperture problem encountered by slow-moving targets and obtain an incomplete structure, making the practical application of satellite video more difficult.

Convolutional neural networks (CNNs) have a superior feature extraction and learning capability [19]. As such, they are a powerful way to extract low-, mid-, and high-level feature representations from images and videos. CNNs can thus perform well in solving complex background changes and background displacement problems. Many studies have applied CNNs to video object segmentation and achieved good results [20]–[23]. These methods pay attention to the motion state of a specific single target, rather than all the moving targets, which is important for military applications and ocean monitoring. The mainstream video object segmentation algorithms mainly focus on learning appearance information, motion information, or a combination of both. The one-shot video object segmentation (OSVOS) method [20] utilizes an offline training approach to generate a general model for the foreground and background segmentation and obtains the segmentation results for a single target through online fine-tuning. However, each frame of the video sequence is segmented individually, and the temporal information is not considered in OSVOS. In order to fully exploit the spatio-temporal information in the large amount of redundant data in satellite videos, many models integrate the segmentation branch and the optical flow branch into a unified network [21], [22]. However, as the optical flow branch needs the optical flow image as input, which is generated and predicted by FlowNet [24], [25], this requires a large amount of extra human effort. Furthermore, the obtained optical flow image may not be accurate enough to guide the object segmentation, due to the relatively low resolution of the satellite data. In order to avoid the generation of optical flow images, some studies have combined target tracking with video object segmentation to model the pixel consistency across time. The region proposal is obtained through object tracking, and the region proposal is segmented to make better use of the consistency relationship between adjacent frames [26]. In addition, a deep Siamese

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encoder–decoder network has been used to take advantage of the mask propagation, which can help to reduce interference and improve the accuracy of the object segmentation [27].

In this article, to efficiently exploit the spatio-temporal information in satellite video and achieve end-to-end SVOS, a spatio-temporal dual-branch network is proposed for the SVOS task. In the proposed dual-branch network, the segmentation task is achieved by two branches, in which the temporal coherence branch aims to mine the spatio-temporal information in the satellite video, and the spatial segmentation branch is used to realize the end-to-end SVOS task. In the temporal coherence branch, predictive feature learning module is used to predict the future video frames through Wasserstein GAN (WGAN) so as to exploit spatio-temporal information from unlabeled satellite video data. For the given successive frames of a satellite video sequence, spatio-temporal information is exploited by predicting the variable and dynamic information of the next frame in adversarial manner. In the spatial segmentation branch, the spatio-temporal features predicted from the temporal coherence branch are concatenated to achieve end-to-end segmentation.

The contributions of this article are as follows.

1) A spatio-temporal dual-branch network is proposed for the SVOS task. In the temporal coherence branch, a predictive feature learning model achieved by WGAN is used to learn the spatio-temporal characteristics from unlabeled satellite video data, thus avoiding complex intermediate processes such as optical flow generation.

2) In the spatial segmentation branch, the fully convolutional network (FCN) is aligned with the spatio-temporal features and applied for end-to-end object segmentation, which avoids the need for hand-crafted features and adopts cross-layer feature fusion to maintain the low-level features. Each of the branch in the network can acquire useful information from the other branch through back-propagation.

3) To solve the problem of the highly unbalanced segmentation in satellite video, boundary loss is used instead of regional loss to achieve the network training, which makes the network pay more attention to the foreground.

The rest of this article is organized as follows. The related works and research foundations of video object segmentation are introduced in Section II. The proposed method is described in detail in Section III, which is followed by the experimental evaluation in Section IV. The parameter analysis of the proposed model and the ablation study are described in Section V. Finally, our conclusions and suggestions for future work are given in Section VI.

II. Related Work

The video object segmentation models can be divided into three categories, based on the degree of human involvement, namely supervised, unsupervised, and semi-supervised. This article mainly focuses on the semi-supervised video object segmentation task, which is aimed at separating an object from the background in a video after having been given the mask of the first frame. Most methods for semi-supervised video object segmentation are based on a temporal consistency approach to propagate the initial mask into the following frames. To this end, graphical models [28]–[30] based on energy function optimization are generally used in the semi-supervised video object segmentation methods. Furthermore, a fully connected graph model [31] can also be used for the long-range spatio-temporal connection. In recent years, video object segmentation based on deep learning has attracted great attention and achieved good results on open datasets. The CNN-based models are mainly aimed at the appearance information and motion information. OSVOS is the representative method of the appearance-based feature learning models, which conducts offline training of a general segmentation model by learning the appearance features, and realizes segmentation of the specified object through online fine-tuning.

The OSVOS architecture consists of a foreground branch and a contour branch. The foreground branch is used to obtain the specific targets, while the contour branch extracts the fine contours of the target. The final object segmentation result can then be obtained by superimposing the two branches. An FCN is utilized for the binary classification task of separating the foreground objects from the background. The final fully connected layer of the classification network is replaced by a convolutional layer, and cross-layer feature fusion is introduced to realize an end-to-end image segmentation network that can accept an input of any size.

The pixel-level cross-entropy loss function for binary classification is defined as

$$\mathcal{L}(\mathbf{w}) = \sum_{j} \log P(y_j = 1|X; \mathbf{w}) + (1 - y_j) \times \log(1 - P(y_i = 1|; \mathbf{w})) \quad (1)$$

where W represents the standard trainable parameters of the CNN, X is the input image, and $y_j \in 0, 1$. $P(\cdot)$ is obtained by applying a sigmoid function to the activation of the final layer.

As OSVOS takes no account of the temporal information, it is not limited by a particular range of motion. However, since most objects in video are in a state of motion compared with still images, motion information is important for video object segmentation. To this end, a lot of research has focused on how to combine appearance information and motion information effectively. A common solution is to use optical flow to obtain the moving region, to maintain temporal consistency. However, optical flow requires a lot of human effort and is not always effective. As a result, an end-to-end sequential learning framework to explore the spatio-temporal features for video segmentation is urgently needed. For example, a sequence-to-sequence network using a convolutional long short-term memory (LSTM) architecture was proposed by Xu *et al.* [32] to fully exploit the long-term spatio-temporal information for video object segmentation by learning the long-term spatio-temporal features directly from the training data in an end-to-end manner. In addition, a novel GAN architecture was proposed by Caelles *et al.* [33], which consists of two "critics" that enforce spatio-temporal consistency to learn spatio-temporal object models over finite temporal windows. It obtains the mask

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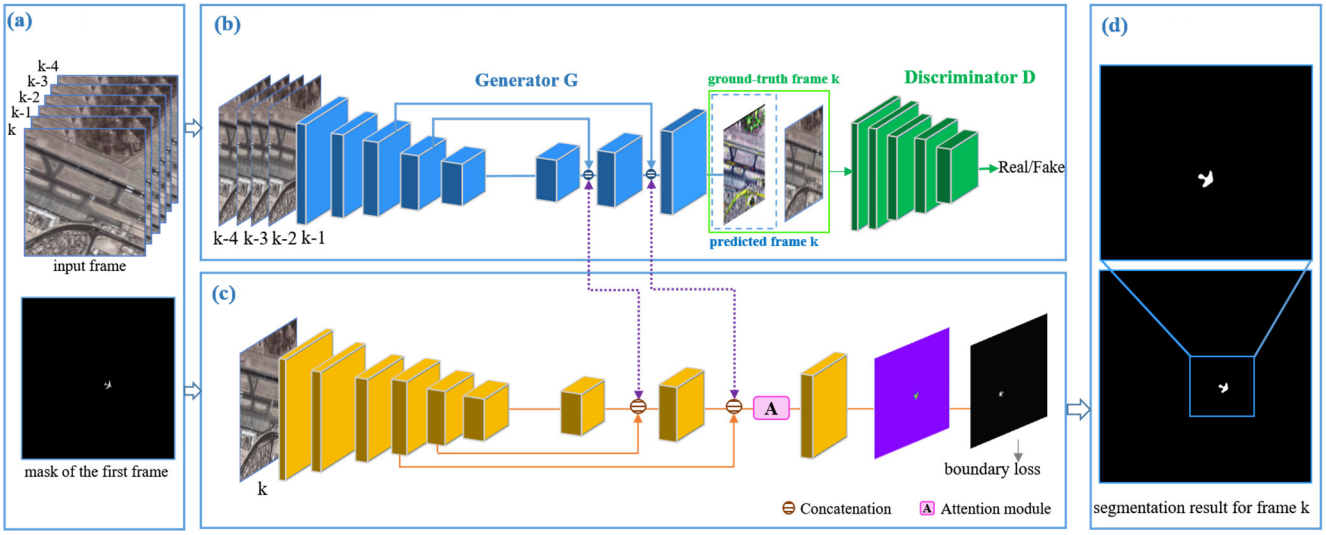


Fig. 2. Proposed model based on a spatio-temporal dual-branch network for satellite video object segmentation. (a) input. (b) temporal coherence branch. (c) spatial segmentation branch. (d) output.

through the mask estimation of the last frame, the current frame, and reference, in which the temporal information is obtained directly through the segmentation result of the last frame.

All of the methods mentioned above for video object segmentation are supervised learning-based methods. However, it is well known that a large amount of training data is needed in the field of supervised learning. Considering the severe burden of human annotation, especially in video, self-supervised learning utilizes prior information from the training data, without manual tags. Differing from supervised learning, which relies on manual annotation of the training data labels, self-supervised learning utilizes the prior knowledge from the training data to provide supervision of the learning representation. Learning in a self-supervised manner unlocks the potential of vast amounts of unlabeled data. A common process for self-supervised learning with video is to train the pretext task model using an unlabeled video dataset, and the intermediate feature layer of the model is provided for the downstream task (e.g., action classification, segmentation, or target tracking) for fine-tuning. The studies of self-supervised learning have made some progress in the field of video. For example, a self-supervised learning visual representation method was proposed by Wang and Gupta [34] through tracking moving objects in the video, in which the visual feature representation is learned by using the similarity of the object tracking frame in different frames, so as to obtain the tracking boxes through self-supervised learning with a large number of unlabeled videos.

Video colorization is the task of coloring grayscale frames into color frames. The consistency of the colors in the video can also be used for the pretext task of self-supervised learning. The video colorization problem was solved by Vondrick *et al.* [35] by learning how to copy colors from reference frames. More recently, self-supervised learning research has also been extended to the video prediction domain.

III. Proposed Framework

The state-of-the-art SVOS methods mainly focus on unsupervised moving object detection without category priors or high-level information, which requires complex post-processing operations. In this article, to solve the above problems, a dual-branch network is proposed to combine the temporal coherence branch and spatial segmentation branch to achieve end-to-end satellite video object segmentation. Differing from the satellite video moving object detection and object tracking tasks, the proposed method is aimed at acquiring complete contours and the location of a single shot with temporal consistency, which is of great significance for the applications of military security and maritime monitoring. The details of the proposed method are described below.

A. Spatio-Temporal Dual-Branch Network for SVOS

Given the satellite video sequence $X = \{X_1, \ldots, X_i, \ldots\}$ and one shot for the target to be segmented in the initial frame, the segmentation result $Y = \{Y_1, \ldots, Y_i, \ldots\}$ is directly obtained through the end-to-end spatio-temporal dual-branch network. In order to maintain the temporal smoothness of the segmentation results, at time t, the previous k frames, namely $X_{t-k}, \ldots, X_{t-1}$ and the current frame X_t, are taken to predict the segmentation results of the current frame Y_t, where k is generally taken as 3, 4, 5, and so on. As shown in Fig. 2, the proposed model is composed of two branches, namely the temporal coherence branch and the spatial segmentation branch. The temporal coherence branch is mainly aimed at future frame prediction by the use of the given past frames [36]–[38] and learns the spatio-temporal features through the predictive feature learning approach, which captures the dynamic appearance and motion cues of the satellite video sequence, instead of using optical flow. At the same time, the spatial segmentation branch is based on an FCN architecture and is used to separate the time-constrained objects from the temporal coherence branch. In the proposed architecture,

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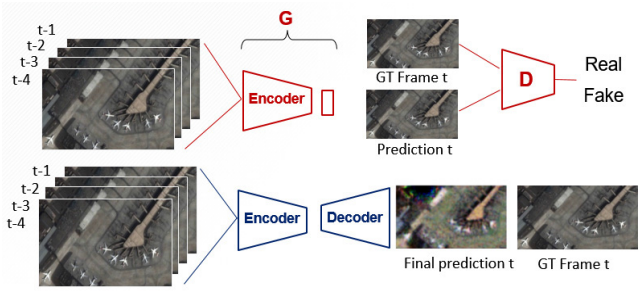


Fig. 3. Framework of predictive feature learning for satellite video.

features acquired by predictive feature learning are used to obtain temporal information and guide segmentation without interfering with the main structure of segmentation. The two branches are described in detail below.

B. Temporal Coherence Branch Network

The large imaging range of satellite video causes a lot of targets to have similar appearances and results in complex background interference. For example, the aircraft in a satellite video sequence may comprise multiple aircraft with similar appearances. The main challenge in SVOS is how to utilize the temporal context information to enhance the smoothness and temporal consistency of the segmentation results. In order to reduce the human effort required for annotation, a predictive feature learning architecture based on self-supervised learning is adopted as the basic structure of the temporal coherence branch. For not requiring time-intensive video annotations, which is too expensive for satellite video, the versatility of predictive feature learning module allows the exchange of static image segmentation and future segmentation prediction in different architectures. The dynamic features acquired by predictive feature learning can also provide prior information for further video object tracking task and improve the drift problem. The pretext task in the temporal coherence branch is to predict the future frames from the previous frames by taking advantage of the continuity of the video. By extracting multi-level spatio-temporal information through the CNN, the abundant spatio-temporal information in the unlabeled satellite video is fully mined and utilized to guide the object segmentation of the spatial segmentation branch.

The predictive feature learning involves learning the intrinsic representation of the video through massive unlabeled videos and has become a promising approach in unsupervised representation learning. Specifically, given a continuous video frame sequence $X_1, X_2, \ldots, X_n$, the predictive feature learning architecture involves constructing a model that accurately generates the subsequent frames $X_{n+1}, X_{n+2}, \ldots, X_{n+t}$ (where t is the number of frames to be predicted). Or, given a sequence in which X_n $(1 < n < N)$ is missing, the model can infer the missing frame by interpolation. In order to predict the real pixel values in future frames, video prediction based on predictive feature learning captures the pixel-level appearance and motion changes so that the pixel values from the previous frame can be passed into the new frame.

The predictive feature learning module is composed of the WGAN, which consists of a generator G and a discriminator D, as shown in Fig. 3. Compared with ordinary GANs [39], the WGAN has the advantages of more stable training and diverse generated samples. At time t, the previous four frames are input for the predictive feature learning module. The generator generates the future frame $G(X_t)$ for the previous frame X_t, while the discriminator D fights against G by identifying the predicted frame $G(X_i)$ and the real frame X_i. In the generator G, an encoder architecture is used to map the input satellite video sequence to the spatio-temporal features and the subsequent output layer to generate the RGB value of the predicted frame through the spatio-temporal features. In the temporal branch, the predicted frames are trained to be closer to the real frame by training D and G in an adversarial manner. The parameters to be trained for D and G are expressed as W_D and W_G, respectively. The goal of training D is to minimize the L_D loss when the generator G is fixed

$$L_D = -\left(1 - D\left(\hat{X}_t\right)\right) - D(X_t) \tag{2}$$

where $X_t{}^{\wedge} = G(\{X_{t-i}\}_{i=1}^{\delta})$ is the generated frame from G based on the previous δ frames, and X_t is the real video frame.

To obtain high-quality spatio-temporal features and predicted frames, the above loss is minimized to make D distinguish between predicted frame G (X_i) and actual frame X_i, which forces G to predict future frames with a higher quality. To optimize the generator, only the adversarial loss for the generator G is generally minimized, which makes G always produce samples that "confuse" D, rather than approximating the true result. In this case, the generator G will produce other samples that deceive D, thus making D unstable.

In order to make D differentiate between real and "confusing" samples more correctly, a combination of adversarial loss and reconstruction loss is used to train the generator, which balances the sharp predictions due to the adversarial principle and similarity with the ground truth brought by the second term, through the mean of parameter λ_{adv}. The generator G generates a video frame that approximates the real one. When the parameter W_D of discriminator D is fixed, the generator G is trained to minimize the joint loss function L_G, which is formulated as

$$L_G = l_{\text{rec}} + \lambda_{\text{adv}} l_{\text{adv}} \tag{3}$$

where

$$l_{\text{rec}} = \left\| X_j - \hat{X}_j \right\|_2 \tag{4}$$

$$l_{\text{adv}} = -D\left(G\left(\hat{P}_i^S\right)\right). \tag{5}$$

By minimizing the combined reconstruction loss and adversarial loss, the generator G can predict the frames that look similar to the corresponding real frames, which is enough to fool the powerful competitor D. In the process of generating the predicted frames, the previous frames are input to predict the future frames, rather than random noise, as in the general GANs. This allows the predicted frame to maintain temporal consistency well. In this case, the temporal coherence branch is able to learn the representation of the implicit temporal cues, which is needed in the SVOS problem. By learning

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the dynamic information through predictive feature learning in the satellite video, the robust and generalized spatio-temporal characteristics obtained from the satellite video can be utilized to generate the real predicted frames. In the spatial segmentation branch, the spatio-temporal information guides and improves the segmentation results.

C. Spatial Segmentation Branch Network

The traditional CNN-based method for image segmentation generally needs a large sliding window, where each window needs space to save the features and identify categories, making the storage cost larger. In addition, as a result of the use of the fully connected structure in the CNN-based models, the storage space of the last few layers increases nearly exponentially, which results in a low computational efficiency. In order to solve the above problems in the image segmentation branch, the FCN architecture is adopted for the image segmentation branch. By modifying the backbone network, the intensive image segmentation prediction is improved for accurate positioning. By removing the fully connected layer required for classification, the modified network can perform image-to-image inference, enabling the segmentation branch to perform effective foreground and background classification for each pixel. The FCN architecture can accept input images of any size, without requiring that all the training images and test images have the same size. The FCN is efficient as it avoids the duplicate storage and convolution calculation caused by the use of image patches. ResNet-101 is used as the backbone network for the FCN, and the convolutional layer in the last two residual blocks (Res4 and Res5) is replaced by a dilated convolutional layer with a stride size of 1. Transpose convolution is applied three times successively for the feature maps to obtain the segmentation image. During the process of transpose convolution, the spatio-temporal characteristics acquired by the temporal coherence branch are input in order to allow the multi-scale feature maps to be obtained by transpose convolution, which means that the segmentation results are constrained with temporal consistency. The multi-scale feature map is then concatenated and the 3×3 convolutional layer is used to obtain the final end-to-end prediction result. In the entire segmentation module, both the spatial segmentation branch and temporal consistency branch can obtain useful information from the other branch through back-propagation. In this way, the spatial segmentation branch can receive useful temporal information from the temporal coherence branch, and the temporal coherence branch can learn more efficient spatio-temporal features for precise segmentation in satellite video.

In the spatial segmentation branch, boundary loss is used instead of the regional loss function to achieve the network training, to solve the problem of the training instability caused by the highly unbalanced segmentation in satellite video. Dice loss is commonly used to solve the problem of the imbalance between positive and negative samples [40]. The definition of dice loss is as follows:

$$L_{\text{dice}} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \tag{6}$$

where $|X \cap Y|$ is the intersection between X and Y, and $|X|$ and $|Y|$ represent the number of elements of X and Y, respectively.

Dice loss is a region-based loss function, in which the loss of the current pixel is not only related to the predicted value of the current pixel, but is also related to the value of the other points in the relevant area. Similar to dice loss, boundary loss can be used to minimize the mismatch between the segmentation result and the ground truth. The main difference between these loss functions lies in the way that the weights are calculated. Boundary loss was first proposed at the Medical Imaging with Deep Learning (MIDL) conference in 2019 [41] and is used for the highly unbalanced segmentation task in medical images. Boundary loss takes the form of a distance metric on the space of the contours rather than regions. It uses integrals over the boundary between regions rather than unbalanced integrals over regions, which can alleviate the difficulty of region loss in the context of highly unbalanced segmentation tasks.

Considering the nonsymmetric L_2 distance for the space of the contours, the formula for evaluating the change between the two adjacent boundaries ∂S and ∂G is as follows:

$$\text{Dist}(\partial G, \partial S) = \int_{\partial G} \| y_{\partial S}(p) - p \|^2 dp \tag{7}$$

where $p \in \Omega$ is a point on boundary ∂G. $y_{\partial S}(p)$ denotes the corresponding point on boundary ∂S, along the direction normal to ∂G, which means that $y_{\partial S}(p)$ is the intersection of ∂S, and the line that is normal to ∂G at p. $||\cdot||$ denotes the L_2 norm.

Boundary loss represents the differential boundary change in the above formula by an integral approach, which represents the boundary change as a regional integral and avoids completely local differential computations involving contour points. The formula for the boundary loss is as follows:

$$L_{\text{BD}} = 2 \int_{\Delta S} \|q - z\partial G(q)\| dq \tag{8}$$

where ΔS denotes the region between the two contours, D_G: $\Omega \rightarrow \mathbb{R}^+$ is a distance map with respect to boundary ∂G, and $D_G(q)$ evaluates the distance between point $q \in \Omega$ and the nearest point $z_{\partial G}(q)$ on contour ∂G.

The boundary loss uses the distance metric formed on the contour space to calculate the distance between the segmentation result and the ground truth. By giving greater punishment weights to the false segmentations which are far away from the ground truth, the network training pays more attention to the regions to be segmented and the regions close by, while less attention is paid to the background regions that are far away from the segmentation. This can help to tackle the unbalanced segmentation problem in satellite video.

The loss function of the generator and discriminator in the predictive feature learning module in the temporal coherence branch and the segmentation loss function in the spatial segmentation branch are combined to form the overall loss function for the proposed satellite video target segmentation model. The combined loss function is defined as

$$L_{\text{combine}} = \arg\min(\lambda_G L_G + \lambda_D L_D + \lambda_{\text{BD}} L_{\text{BD}}) \tag{9}$$

where λ_G, λ_D, and λ_{BD} are hyper-parameters that control the relative importance of each loss term.

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D. Network Implementation

The network implementation of the proposed model includes pre-training, offline training, and online fine-tuning. Both the offline training and online training are critical for the network implementation. The temporal coherence branch is pre-trained by surveillance video data, and the spatial segmentation branch is pre-trained by ImageNet images. The implementation is based on the public PyTorch platform. At the pre-training stage, the temporal coherence branch and the spatial segmentation branch are pre-trained individually. Since the imaging approach and calibration of satellite video are different from the imaging approach and calibration of surveillance video, it is not possible to train the CNN structure with surveillance video. Therefore, the joint training phase of the model mainly relies on satellite video data. Due to the lack of datasets for the SVOS task, the *densely annotated satellite video object segmentation* (DAVOS) dataset was built with fine annotation for the SVOS task. A detailed description of the DAVOS dataset is given in Section IV. The temporal coherence features are also concatenated to integrate temporal constraints. The spatial features are concatenated to the temporal coherence branch to improve the predicted features in the same way. The two branches are then joined together for offline training and iterative updating by the DAVOS dataset. For a given object mask to be segmented in satellite video, the joint models are fine-tuned on each satellite video sequence for the online training.

In this study, the temporal coherence branch was pre-trained by the training set and validation set of the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2015 object detection from video (VID) dataset [42]. This dataset contains a total of 4417 video clips, including 3862 video clips in the training set and 555 video clips in the validation set. In the temporal coherence branch network, the encoder structure is mainly used for mapping the input video sequence to the spatio-temporal feature map. The selected backbone network was Res101, which was initialized by the pre-trained model on the ILSVRC object classification/localization (CLS-LOC) dataset [50]. The backbone network of discriminator D was GoogLeNet, which was initialized by the pre-trained model on the ILSVRC CLS-LOC dataset [50]. In order to enhance the model training, all the frames in the video clip were randomly flipped horizontally to increase the training data. The batch size was set to 5, and the Adam optimization algorithm was used to train the model. The input video frame number was set to 4. During the training, the trade-off weight λ_{adv} was set to 0.001. The learning rates of generator G and discriminator D were set to 10^{-7} and 10^{-4}, respectively. In the spatial segmentation branch, the PASCAL VOC 2012 segmentation dataset was used for the pre-training. The PASCAL VOC 2012 dataset contains 11 355 images. The images were randomly flipped and rotated to increase the training data. Stochastic gradient descent (SGD) was used to optimize the model, while the batch size was set to 10 and the learning rate was 10^{-3}. In addition, boundary loss was applied to the multi-scale prediction to calculate the total training loss.

After pre-training of the temporal coherence branch and spatial segmentation branch, the joint model was trained on the DAVOS dataset with six video clips. Specifically, the temporal coherence branch and spatial segmentation branch are iterative training processes. For this iterative learning process, each branch in the network can learn useful information from the other branch through back-propagation. In this way, the spatial segmentation branch can learn useful spatial information from the temporal coherence branch, and the temporal coherence branch can learn more efficient spatio-temporal features for precise segmentation. The weights of the spatial segmentation branch is fixed. When the temporal coherence branch was optimized, the learning rates of the generator G and discriminator D were 10^{-8} and 10^{-4}, respectively. In the training process, the batch size was set to 1, and the RMSProp optimizer [43] was used to optimize the weights of the temporal coherence branch. The clipping parameters for the discriminator D weights were set to -0.01 and 0.01. For the training of the spatial segmentation branch, the weights of the temporal coherence branch were also fixed, and the SGD algorithm with a learning rate of 10^{-4} was used to update the weights of the spatial segmentation branch.

Finally, the network was fine-tuned during the online training for a few iterations on the particular instance we wanted to split to obtain a more precise object segmentation result. Three satellite videos from the DAVOS dataset were used for the online testing. Each mask in the first frame was augmented to generate multiple training samples, to increase the diversity of the test set. In the training process, the SGD algorithm was used for the online training, with the learning rate set to 10^{-4} and the batch size set to 1.

TABLE I
DETAILS OF THE DAVOS DATASET

DAVOS benchmark dataset	Train	Val	test	Total
Number of sequences	10	4	3	17
Number of frames	2250	1215	792	4257
Mean number of frames per sequence	225	303.8	264	250.4

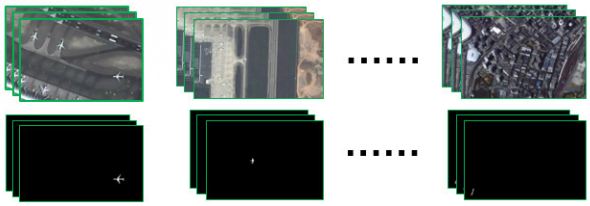


Fig. 4. Example frames of the DAVOS dataset with precise annotation for one-shot satellite video object segmentation.

IV. EXPERIMENTS AND ANALYSIS

A. Experimental Setups

1) Dataset: The DAVOS dataset has one annotated object in each scene for each satellite video sequence. A detailed description of the DAVOS dataset is given in Table I. Some example frames from the DAVOS dataset are shown in Fig. 4. A dataset with a large volume is needed to ensure the diversity of the content and prevent overfitting of the training. To this end, the DAVOS dataset comprises a total of 17 sequences,

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with the data made up of SkySat, UrtheCast, and Jilin-1 satellite videos with an average spatial resolution of 1 m. The DAVOS dataset is split into three parts, with ten satellite video sequences as the training sets, four video sequences as the validation sets, and three video sequences as the test sets. The satellite video sequences have a short temporal extent, with about 300 frames per video. There is only a single annotated object in each frame, and all the videos were captured at 24 frames/s. Differing from the video object segmentation datasets for surveillance video, the scenes for satellite videos are generally large and have a large number of targets, often including small moving vehicles and other small targets. However, what people commonly pay attention to in satellite video is certain targets with typical characteristics, such as military targets. According to the actual requirements of the satellite video, two common and typical categories of aircraft and trains are used for the fine annotation. The annotations for each type of target are evenly distributed in the dataset.

2) Comparison Methods: In the following, the experimental results of the proposed method and the comparative methods obtained with the DAVOS dataset are analyzed in detail, including both a qualitative analysis and a quantitative comparison of the results. Four unsupervised foreground detection methods were compared in the experiments, that is, the average background subtraction (ABS) method [19], the three-frame-difference (TFD) method [44], ViBe [11], and SAmple CONsensus (SACON) [45], in addition to the semi-supervised models of OSVOS [20], the space-time memory (STM) network [46], SiamMask [47], and FaSTGAN [33].

3) Evaluation Metrics: For the SVOS task, three metric indices are used to evaluate the performance of the model, namely the region similarity J, the contour accuracy F, and the temporal stability T. In a supervised evaluation framework, the fitting degree of M and G should be measured for the given background truth mask G and the output segmentation M on a specific frame. To this end, the region similarity J is used to measure the matching degree of the predicted mask and true mask, while F is used to measure the precision of the contours of the predicted mask. In addition, as satellite video extends the dimension of static imagery to the temporal dimension, the temporal stability of the results should also be considered in the segmentation results. The three evaluation indicators are described in detail below.

In order to measure the similarity between the segmentation result and the ground truth, the region similarity J is used to define the intersection-over-union of the segmentation result and the real mask. The region similarity J provides intuitive, scale-invariant information about the number of error-marked pixels. Given an output segmentation result M and the corresponding ground-truth mask G, the region similarity J is defined as

$$J = \frac{|M \cap G|}{|M \cup G|}. \tag{10}$$

The precision of the acquired contours is also important for object segmentation. As mentioned above, the output segmentation result M can be interpreted as a set of closed contours $c(M)$ delimiting the spatial extent of the mask. The contours of the ground-truth mask are interpreted as $c(G)$ in the same way. The contour-based precision P_c and recall R_c between the contour points of $c(M)$ and $c(G)$ are calculated via bipartite graph matching. The performance of P_c and R_c is compromised by the trade-off matric, namely the contour accuracy F. The contour accuracy F is defined as

$$F = \frac{|2P_cR_c|}{P_c + R_c}. \tag{11}$$

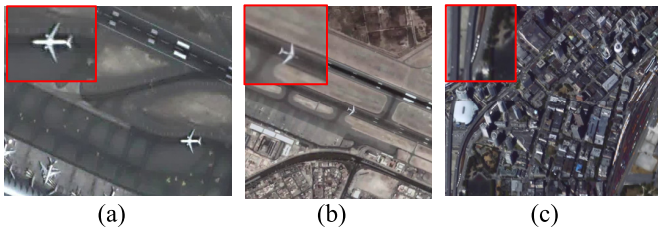


Fig. 5. First frames of the three test datasets. The red rectangles show the target that needed to be segmented in each satellite video scene. (a) First plane sequence. (b) Second plane sequence. (c) Train sequence.

Differing from static imagery, the temporal stability of the result is an important indicator for SVOS. As the evolution of the object shapes is an important clue for recognition, unstable boundaries are not acceptable in video object tracking. Therefore, temporal stability measures are used to determine these undesired effects. The temporal stability is obtained by calculating the average cost of deforming the mask from the previous frame to the next frame. This can be obtained by using dynamic time warping (DTW) [48] to minimize the shape context descriptor (SCD) [49] distance between the matching points of the consequent frames.

B. Experiment 1: First Plane Sequence

To verify the effectiveness of the proposed method on satellite video sequences, three satellite video sequences were used to realize the SVOS task. As shown in Fig. 5(a), the first satellite video is a plane sequence, with a target size of about 65 × 65 pixels. Fig. 6(a) shows the segmentation results for the first plane scene. Fig. 6 (b)–(e) shows the segmentation results of the unsupervised SVOS models of ABS, TFD, ViBe, and SACON, respectively. Fig. 6(f)–(i) shows the experimental results of the semi-supervised OSVOS, STM, SiamMask, FaSTGAN, and the proposed method, respectively.

In the first plane scene, some planes are highly similar to the target to be segmented, which can easily interfere with the segmentation results. The background subtraction models are heavily dependent on the selection of the segmentation threshold and the quality of the background model, which are prone to causing foreground aperture problem. Although the ABS and SACON methods can clearly identify the foreground object, they only acquire the general outline of the target. As there are empty holes inside the foreground, the video object segmentation results obtained by background subtraction are incomplete. In this case, ViBe only uses a single frame for the background modeling and acquires a more complete target, but there is still noise in the segmentation results.

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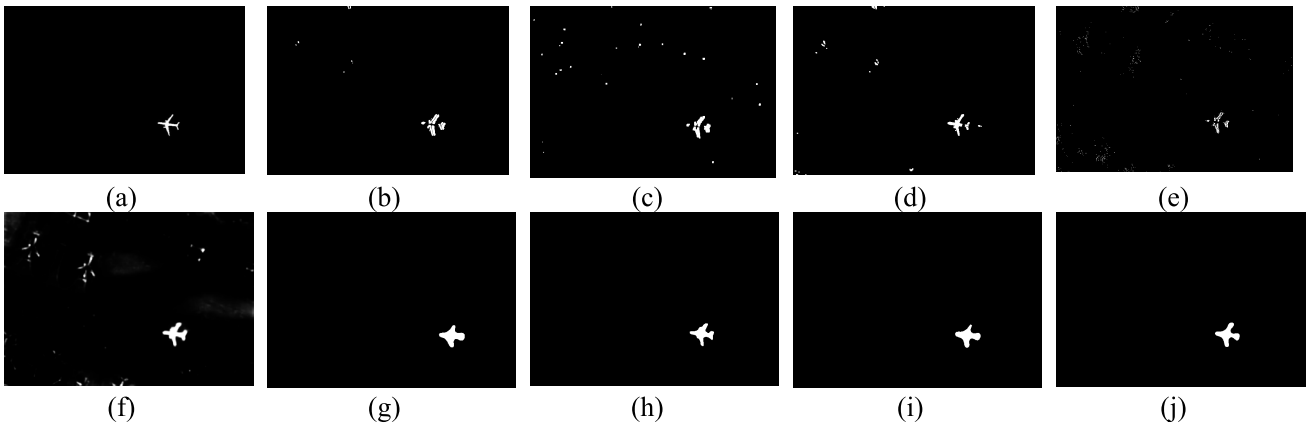


Fig. 6. Segmentation results for the 100th frame of the first plane sequence. (a)–(j) Ground truth and the segmentation results of ABS, TFD, ViBe, SACON, OSVOS, STM, SiamMask, FaSTGAN, and the proposed method, respectively.

TABLE II
QUANTITATIVE EVALUATION RESULTS FOR EXPERIMENT 1

Metric	ABS [10]	TFD [41]	ViBe [12]	SACON [45]	OSVOS [20]	STM [46]	SiamMask [47]	FaSTGAN [33]	Proposed
J (↑)	0.3546	0.3095	0.3862	0.1942	0.3851	0.5139	0.4473	0.5091	**0.5566**
F (↑)	0.8720	0.5696	0.7336	0.4430	0.4721	0.9480	0.9943	0.9401	**0.9926**
T (↓)	1.2922	1.3222	0.4791	1.8701	0.1588	0.4045	0.3715	0.3501	0.1682

The TFD method fails to effectively segment the foreground from the dynamic background, due to its sensitivity to noise. The under-segmentation problem is also apparent in the segmented foreground objects, where only the contours of the target are acquired. The four deep-learning methods acquire end-to-end segmentation results while avoiding the complicated intermediate process. It shows that the OSVOS model obtains a more complete outline and internal structure of the segmented target, which solves the foreground aperture problem and fully demonstrates the advantage of the end-to-end segmentation models. However, as the OSVOS does not fully consider the temporal information of the satellite video, some false alarms are generated in the segmentation image, and a large number of false alarms are generated in this complex scene. The STM, SiamMask, and FaSTGAN acquire better visual results for accounting temporal information. Due to strong real-time performance and lack of online fine-tuning process, the contour accuracy of SiamMask results is relatively worse. Fig. 9(d) shows the example of predicted frame for the first plane sequence, and Fig. 9(e) shows the example of the predicted features. By applying the spatio-temporal feature to constrain the segmentation results, the proposed method filters the false alarm targets and interference in the background, thus obtaining robust and accurate segmentation results.

Table II lists the quantitative evaluation results for Experiment 1. The ↑ symbol indicates that the larger the value, the better the segmentation result. The ↓ symbol indicates that the smaller the value, the better the segmentation result. As the imaging range of the satellite video scene is very wide, the target only occupies a very small proportion of the satellite video sequence. As a result, except for the method proposed in this article, the region similarity J values of the compared models are very small and are all less than 0.5. In contrast, the proposed method increases the region similarity J to 0.5494. For the contour accuracy, the proposed method again performs well and outperforms the unsupervised methods and the OSVOS model. In terms of the temporal stability, the four methods based on deep learning perform better than the unsupervised segmentation models. The T index value of the proposed method is 0.2501, which is the second lowest value among all the methods. Table II shows that the three evaluation indicators all confirm that the proposed method constrains the segmentation results by considering the temporal information, thus obtaining a superior segmentation performance.

C. Experiment 2: Second Plane Sequence

As shown in Fig. 5(b), compared with the first plane sequence, the second satellite video sequence has a smaller size of target, with a target size of about 40 × 60 pixels. The second satellite video sequence has less interference from similar targets, but the background interference is more complex and there are more moving targets, such as vehicles.

Fig. 7 shows the segmentation results for the second plane scene. Fig. 9(f) shows the predicted frame for the second plane sequence, and Fig. 9(i) shows the predicted features acquired by the temporal coherence branch. It shows that ABS, TFD, ViBe, and SACON obtain a rough outline of the target, but the foreground aperture are apparent in the segmentation results. TFD and ViBe also show obvious false alarms and noise in the segmentation results. ABS obtains rough target contours, but there is an obvious cavity area, while TFD also obtains rough target contours, plus some false alarms. Similarly, both ViBe and SACON only obtain incomplete target contours. As there is no similar target interference in the

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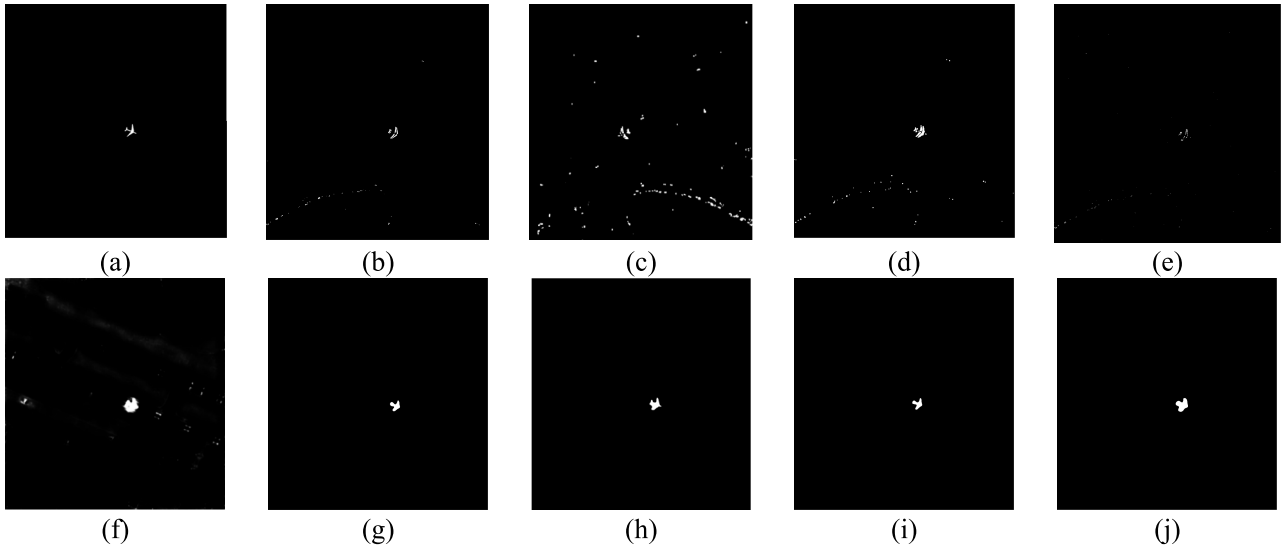


Fig. 7. Segmentation results for the 35th frame of the second plane sequence. (a)–(j) Ground truth and the segmentation results of ABS, TFD, ViBe, SACON, OSVOS, STM, SiamMask, FaSTGAN, and the proposed method, respectively.

TABLE III
QUANTITATIVE EVALUATION RESULTS FOR EXPERIMENT 2

Metric	ABS [10]	TFD [41]	ViBe [12]	SACON [45]	OSVOS [20]	STM [46]	SiamMask [47]	FaSTGAN [33]	Proposed
J (↑)	0.2371	0.1290	0.2669	0.0935	0.3416	0.3980	0.5294	0.5261	0.5319
F (↑)	0.6144	0.2329	0.5514	0.4082	0.5657	0.9231	0.9089	0.9502	0.9691
T (↓)	2.1918	1.4354	0.4358	2.4899	0.1747	0.2684	0.5088	0.3561	0.1830

scenario, OSVOS obtains a good segmentation result, while not considering the temporal information. However, despite the false alarms being restrained, there are fuzzy contour boundaries in the segmentation result of the OSVOS model. Compared with the OSVOS model, the STM, FaSTGAN, and the proposed method obtains a complete boundary and target structure. But for the accuracy of the contour, the FaSTGAN and the proposed method are more accurate.

Table III lists the quantitative evaluation results for Experiment 2, where it can be seen that the results for Experiment 2 are similar to the results for Experiment 1. In terms of the J and F values, the proposed method achieves the best results. As a large number of false alarms are suppressed, the proposed method improves the F index score by about 50% when compared with OSVOS and performs well than other three semi-supervised methods.

D. Experiment 3: Train Sequence

Experiment 3 was aimed at separating the train from the satellite video sequence, for which the target size is about 30×80 pixels. The data used in Experiment 3 were released as part of the 2016 IEEE Data Fusion Competition and were recorded by the UrtheCast satellite. The train shows obvious appearance changes throughout the sequence. Therefore, consideration of the temporal information is more important in this train scene. As shown in Fig. 5(c), the proportional size of the train in the satellite video sequence is very small, and the background is relatively complex. In addition, the scene is very large, and the moving objects in the scene include not only the train, but also moving vehicles and background displacement. The interference caused by the building shadow and the illumination change all make the segmentation relatively difficult.

Fig. 8 shows the segmentation results for the train scene. Fig. 9(e) shows the predicted frame for the train sequence, and Fig. 9(h) shows the predicted features. It shows that there are some cracks and some false alarms in the results obtained by ABS and TFD. The segmentation results acquired by ViBe only include the front and back parts of the train, while SACON acquires fewer false alarms. The four semi-supervised models based on deep learning obtain a complete target counter and structure. As the FCN does not consider the semantic information between targets in the segmentation, OSVOS shows relatively serious background interference in the background area. In contrast, the proposed method can suppress the background interference well because it fully considers the temporal information.

Table IV lists the quantitative evaluation results for Experiment 3. Because the size of the train is relatively small and the SVOS task is more difficult, the region similarity values of most of the models are relatively low, whereas the method proposed in this article achieves a relatively high value. In terms of the F index, the proposed method shows a very significant improvement over the other methods. In terms of the temporal stability, the proposed method also obtains a superior result.

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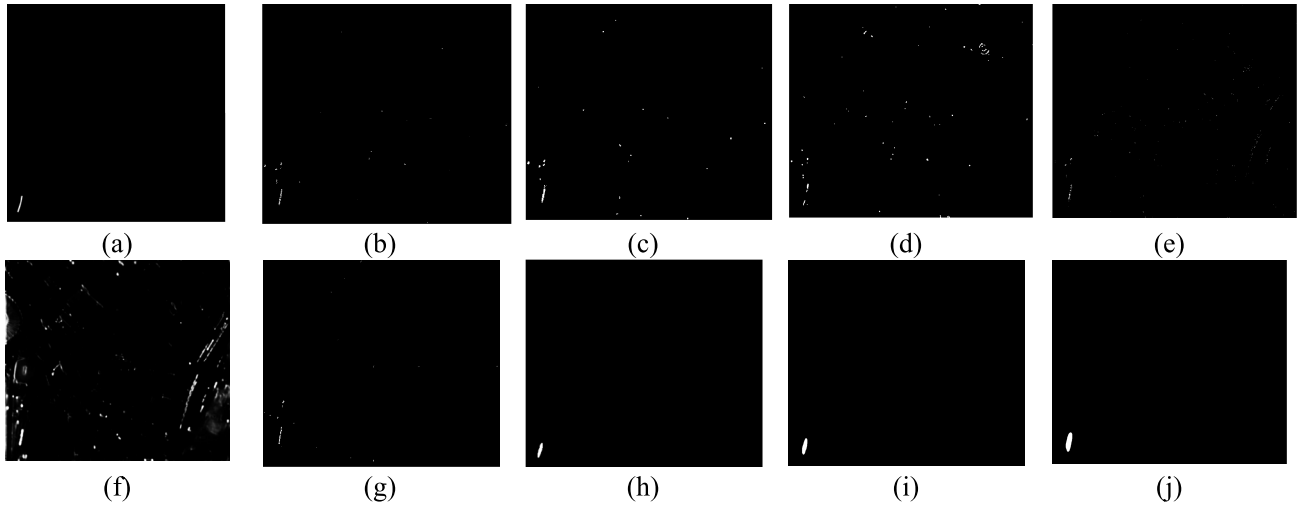


Fig. 8. Segmentation results for the 80th frame of the third train sequence. (a)–(j) Ground truth and the segmentation results of ABS, TFD, ViBe, SACON, OSVOS, STM, SiamMask, FaSTGAN, and the proposed method, respectively.

TABLE IV
QUANTITATIVE EVALUATION RESULTS FOR EXPERIMENT 3

Metric	ABS [10]	TFD [41]	ViBe [12]	SACON [45]	OSVOS [20]	STM [46]	SiamMask [47]	FaSTGAN [33]	Proposed
J (↑)	0.1899	0.2187	0.0816	0.1121	0.0547	0.1980	0.1721	0.3678	0.5361
F (↑)	0.5526	0.4611	0.2161	0.4251	0.0780	0.7891	0.4654	0.8781	0.9982
T (↓)	1.7092	1.3980	0.9056	2.1039	0.2342	0.2671	0.2528	0.4892	0.1554

TABLE V
ABLATION STUDY RESULTS FOR THE TEMPORAL COHERENCE BRANCH

Metric	J	F	T
Temporal	0.53189	0.99003	0.21917
Non-Temporal	0.42183	0.96513	0.60092

V. ABLATION STUDY AND PARAMETER ANALYSIS

In this section, in order to better explain the proposed model, the ablation study for temporal coherence branch, WGAN and adversarial training loss are given in detail. The parameter sensitivity analysis conducted for the segmentation loss and segmentation threshold is also discussed in detail.

A. Ablation Study for the Temporal Coherence Branch

An ablation study for the temporal coherence branch was performed to verify the importance of the temporal information in SVOS. Table V lists the quantitative results of the model considering temporal information and the model not considering temporal information. It can be seen that, when not considering the temporal information, the J index decreases by about 0.11 and the F index by about 0.025. Most notably, in the absence of temporal coherence guidance, the value of the temporal stability T decreases significantly, by 63.53%, which indicates that the temporal information plays a very important role in the stability and temporal consistency of the segmentation results.

B. Sensitivity Analysis for the Proposed Model

In the proposed model, the segmentation loss function and the segmentation threshold are two critical factors. In order to verify the effectiveness of the segmentation loss for the SVOS task and the sensitivity of the segmentation results to the segmentation threshold, the segmentation performance of the three segmentation functions was evaluated under different segmentation thresholds. Fig. 10 shows the evaluation results for the three segmentation loss functions (binary cross-entropy loss, dice loss, and boundary loss) as threshold t changes, for the evaluation indices of J, F, and T.

Fig. 10(a) shows the performance of the three loss functions with regard to the region similarity J with the change of the probability threshold t. When t changes from 0.1 to 0.6, the evaluation results for the dice loss and boundary loss functions are relatively stable, while the binary cross-entropy loss function shows a significant decrease in the value of J when the threshold t is close to 0.45. Overall, the boundary loss function shows the best performance among the three segmentation loss functions. Similarly, for the contour accuracy F, Fig. 10(b) shows that the boundary loss function again performs the best among the three segmentation loss functions, while the binary cross-entropy loss function performs the worst. In terms of the temporal stability T, Fig. 10(c) shows that the temporal stability T decreases with the increase of the probability threshold t. Among the three loss functions, the temporal stability of the boundary loss function is maintained at a relatively stable and high level.

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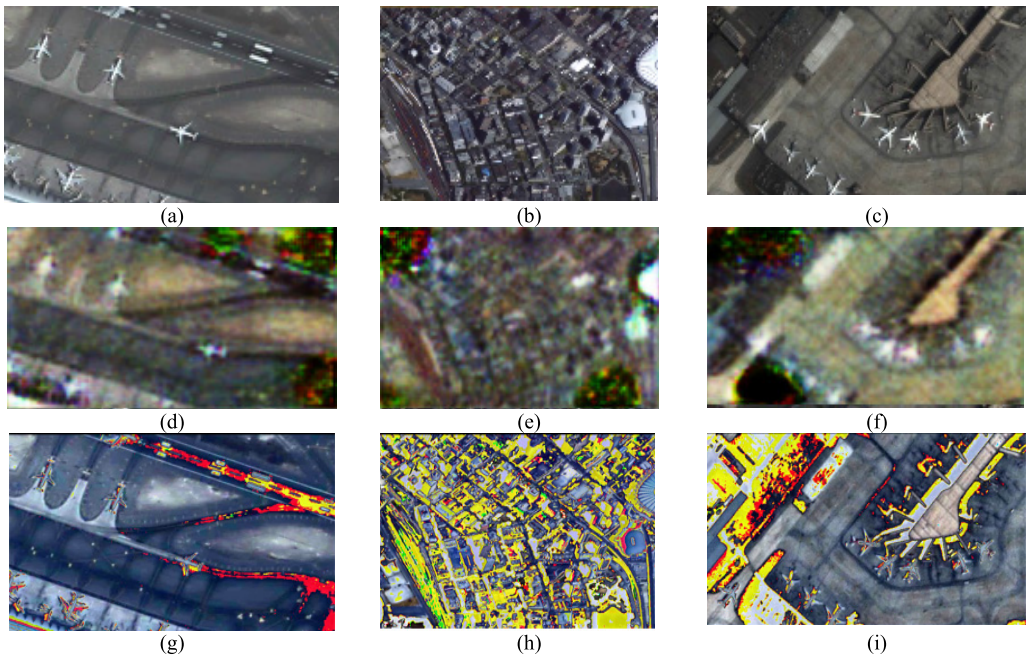


Fig. 9. Examples of predicted satellite video frames. (a)–(c) Three real frames. (d)–(f) Predicted frames. (g)–(i) Features predicted by predictive feature learning.

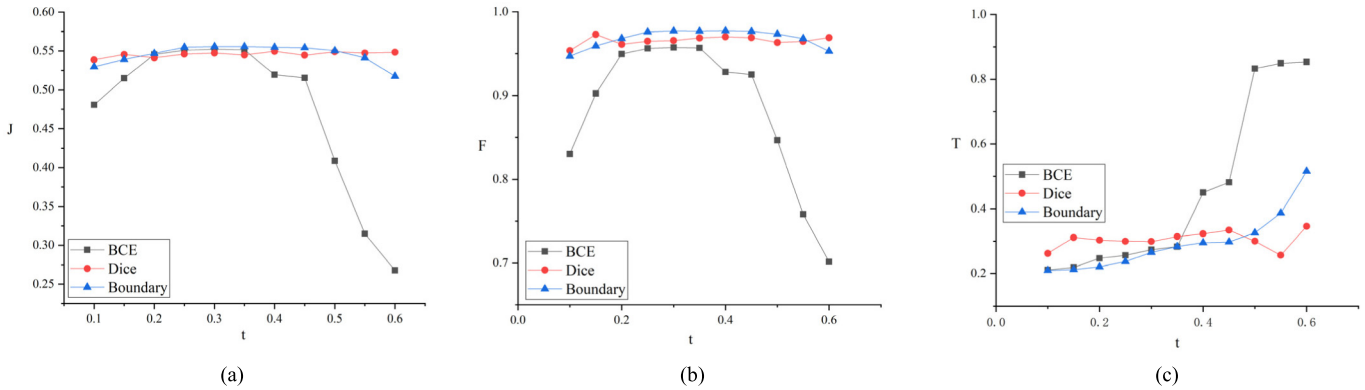


Fig. 10. Parameter sensitivity analysis for the proposed model. (a)–(c) Different performances for evaluation indices J, F, and T for the three segmentation loss functions (binary cross-entropy loss, dice loss, and boundary loss) with the change of the probability threshold t, where the step size of t was set to 0.05.

According to the results of the parameter sensitivity analysis, the boundary loss function shows a good performance for the SVOS task. According to the variation of the boundary loss function for J, F, and T with the change of t, the optimal segmentation threshold was found to be 0.25.

C. Ablation Study for the Adversarial Training

In order to generate more realistic predicted image frames in the predictive feature learning module, the combined L_2 loss and adversarial loss is applied to achieve the network training, so as to avoid the fuzzy prediction problem caused by using L_2 loss exclusively, which is generally related to the inherent uncertainty of video prediction. The GAN generates more accurate and diverse images due to the increase of adversarial training, instead of relying entirely on the paradigm distance between the generated image and the real image for error calculation. To verify the effectiveness of adversarial loss in predictive feature learning, Fig. 11 shows an example of the frame prediction results for the DAVOS dataset. According to the frame prediction results, it can be concluded that the use of exclusively L_2 loss leads to image blurring and the generation of some inaccurate prediction areas. Table VI shows the accuracy evaluation results on three experimental data for the ablation study of adversarial loss. The results show that adversarial loss is worked in the network training of the



Fig. 11. Comparison example of frame prediction with the DAVOS dataset. (a) Ground truth. (b) Frame prediction result by the combined L_2 and adversarial loss. (c) Frame prediction result by exclusively L_2 loss.

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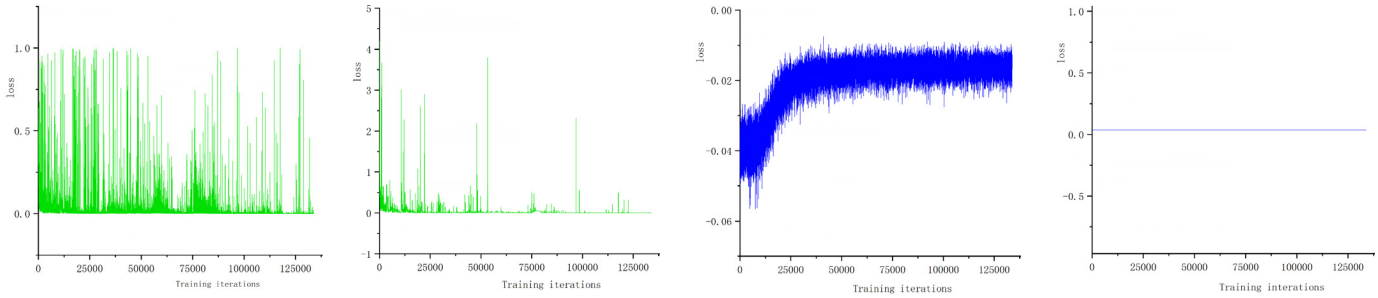


Fig. 12. Comparison of the training loss for the original GAN and the WGAN. (a) Generator loss of GAN. (b) Discriminator loss of GAN. (c) Generator loss of WGAN. (d) Discriminator loss of WGAN.

TABLE VI
ACCURACY RESULTS OF ABLATION STUDY FOR THE ADVERSARIAL LOSS

	Experiment1		Experiment2		Experiment3	
	L_2	L_2+L_{adv}	L_2	L_2+L_{adv}	L_2	L_2+L_{adv}
J	0.54714	0.55660	0.50662	0.54645	0.50809	0.53614
F	0.96824	0.99258	0.92285	0.96907	0.99723	0.99819
T	0.18586	0.16819	0.17611	0.18298	0.14469	0.15544

proposed method. The J and F indicators are higher than using L_2 only. In addition, since WGAN is introduced into the network training, the evaluation result of time stability with adversarial loss remains relatively consistent with that of L_2 loss only. Thus, the above comparison results verify the validity of the adversarial loss function in the predictive feature learning in the proposed model.

D. Ablation Study for the WGAN

The WGAN is used to solve the unstable training problem. By introducing the Wasserstein distance, the WGAN can solve the gradient vanishing problem, as a result of its superior smoothness when compared with Kullback–Leibler divergence and Jensen–Shannon divergence. The WGAN approximates the Wasserstein distance by converting it into a solvable form by mathematical transformation and maximizing it using a discriminator neural network with a limited range of parameters. The Wasserstein distance can be reduced by optimizing the generator under this approximate optimal discriminator, and the generated distribution and the real distribution can be narrowed effectively. Thus, the problem of unstable training with GANs can be solved, and the training degree of the generator and the discriminator need not be carefully balanced. This also solves the collapse mode problem and ensures the diversity of the generated samples. A comparison of the results of the training loss for the original GAN and the WGAN is provided in Fig. 12. As is shown, compared with the original GAN, the loss of the WGAN is relatively stable, on the whole, and the gradient does not disappear in the process of the multiple trainings.

VI. CONCLUSION

In this article, a spatio-temporal dual-branch network with predictive feature learning has been introduced for the SVOS task. The previous research into satellite video has mainly focused on unsupervised foreground target extraction. However, the state-of-the-art algorithms cannot adapt to the complex background in satellite video and obtain an incomplete target structure, resulting in difficulty for the object tracking and further satellite video applications. In order to fully utilize the spatial and temporal information in satellite video, a spatio-temporal dual-branch network for SVOS has been proposed in this article. In the spatial segmentation branch, an FCN is applied to achieve end-to-end object segmentation. To solve the unbalanced foreground and background problem, boundary loss is applied to achieve the network training. In the temporal coherence branch, the predictive feature learning approach from unlabeled satellite video data learns the spatio-temporal information and models the rich content of the variable and dynamic information of the video, which restrains the segmentation results from the spatial segmentation branch. The experimental results obtained in this study fully verified the superior performance of the proposed method for the SVOS task.

However, limited by the spatial resolution of satellite video, the segmentation results did not achieve a very high precision in contour accuracy. Our future work will focus on super-resolution reconstruction for satellite video, to improve the image quality and precision of the contours.

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